

TASK 1 (20p)

Create **rectangle** and **cone** classes derived (public) from **area_cl** given below. rectangle class will be calculate area of rectangle, cylinder class will be calculate area of cylinder ($\pi = 3,14$).

```
#include <iostream>
using namespace std;

class area_cl{
public:
    double height, width;
};
```

Both classes must have **double area (double h, double w)** function. Dimensions will be entered by user. The program you developed should work as in the example below.

Example (working code):

```
What is the dimensions?
4
5
Cone's area is :46.4023590073
Rectangle's area is :20
```

Note 1: width value is diameter for cone

Note 2: Codes must be start with student number, name and surname (an example given below)

```
1 // 20xxxxxxxx Aaaaaa Bbbbbb
```

Note 3: You must check your code for any vulnerabilities. Example:

```
Please enter two numbers :
a
s
You entered characters, please enter numbers
Please enter two numbers :
```

TASK 2 (20p)

Fill the blanks and rewrite the code according to the given output of the code. Assume that all required libraries are included to the code.

Code:

```
using namespace std;
class truck;
class car{
    int passengers, speed;
public:
    car(int p, _____) { passengers = p; speed = s; }
    int sp_greater(truck t);
};
class truck{
    int weight, speed;
public:
    truck(int w, int s) { weight = w; speed = s; }
    _____ int car::sp_greater(truck t);
};
int car::sp_greater(truck t){ return speed - t.speed; }
int main(){
    int t; car c1(2);
    truck t1(20000, 72);
    cout << "Comparing c1 and t1:\n"; t = c1.sp_greater(t1);
    if(t<0) cout << "Truck is faster.\n";
    else if(t==0) cout << "Car and truck speed is the same.\n";
    else cout << "Car is faster.\n";
    return 0;
}
```

Output of the code:

```
Comparing c1 and t1:
Car is faster.
```

TASK 3 (40p)

Answer the questions below. Give a short answer.

- 1- What is the responses of **new** to an allocation error?
- 2- What is difference between **reference** and **pointer**?
- 3- What is the advantages of **reference**?
- 4- What are the differences between **access-types**?
- 5- What requirement/requirements must be met for an **object to be assigned to another object**?
- 6- How should a **function be defined** so that it can use private members of the class it is not a member of?
- 7- What is the differences between **new()** and **malloc()**, **free()** and **delete()** functions?

REPORT RULES (20p)

- Write the subject of the e-mail you will send as **x.Education 20xxxxxxx**.
- Send only one e-mail. If you send more than one e-mail, the first e-mail sent will be evaluated.
- In reports you should add a screenshot of a executed code.
- Reports should be written in **English**. Reports written in other languages will not be **evaluated**.
- You will send your reports to **cucen216@gmail.com**. The deadline for report submission is **9 May at 23:59**. Reports submitted after this date will not be **evaluated**.

If the above mentioned rules are not followed, **your report score will decrease**.